



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO KSS ENTERPRISES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

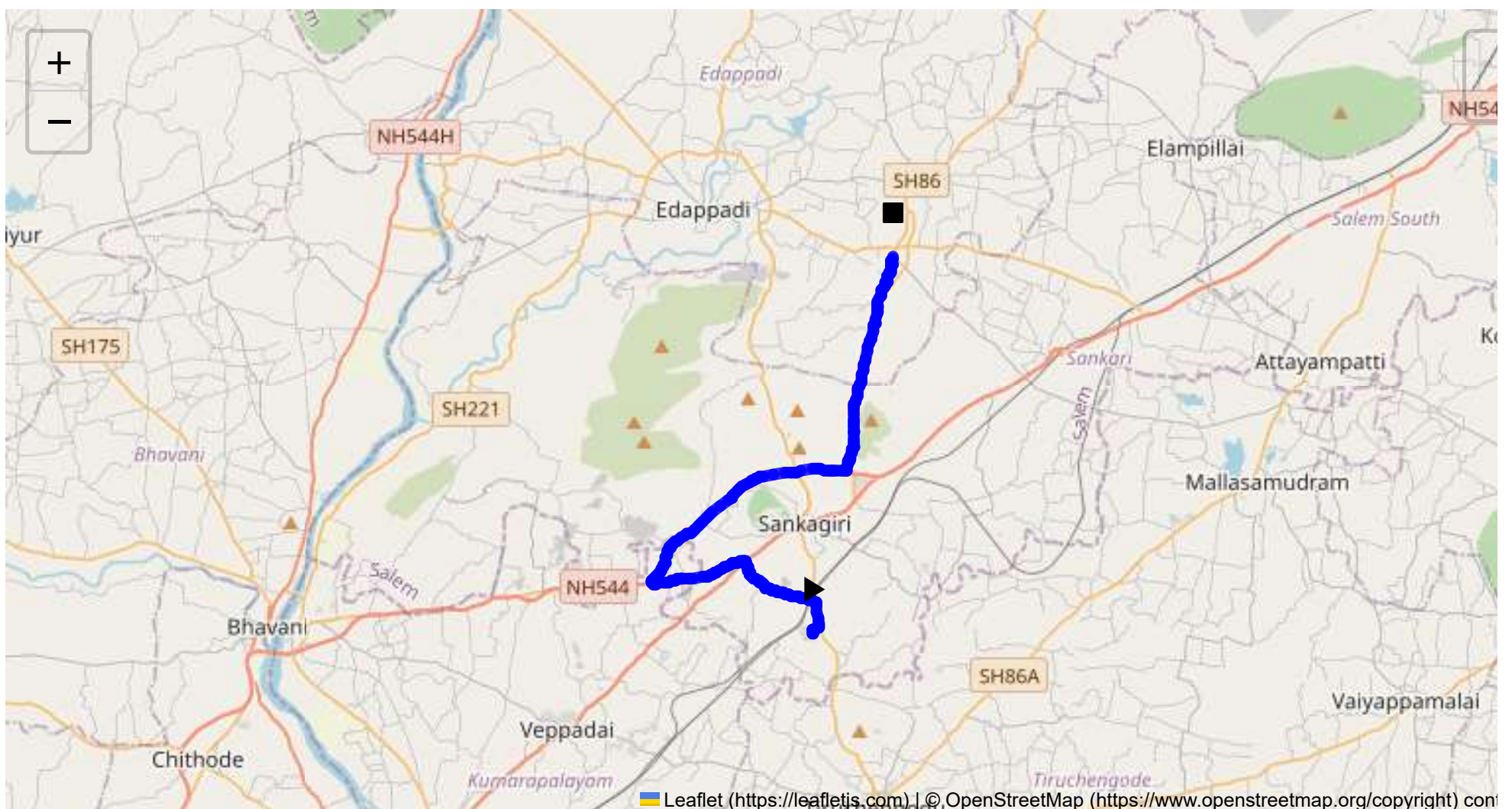
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 26.21 km**  
**Estimated Duration: 0.6 hours**  
**Adjusted Duration (Heavy Vehicle): 0.7 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.56748974, 77.90079449)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to Ground Floor Sankari Main Rd, Konganapuram spans approximately 26.21 kilometers. It includes travel through Puthur, Bhavani Main Road in Padaiveedu, Salem - Kochi Highway in Sangagiri, and State Highway 86 in Erumaipatti. The route passes through both rural and semi-urban areas, with a mix of local roads and a highway.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with relatively high humidity. The monsoon season, typically from June to September, can bring heavy rainfall, leading to potential flooding and road slipperiness. Summers (March to June) can be hot, which might affect road surface conditions. Fog is occasionally noted during early winter mornings which can reduce visibility.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Typically in the morning between 8:00 - 10:00 AM and evening between 5:00 - 7:00 PM.
- **Congestion-Prone Areas:**
  - Intersection points with Salem - Kochi Highway
  - Market areas within villages like Puthur and Sankari Main Road, Konganapuram.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally maintained but variable. Some patchy areas due to heavy vehicle use, particularly on local roads.
- **Infrastructure:** The major highways are well developed, but smaller connecting roads may lack adequate signage and lighting.

## 5. Suggestions for Alternative Routes for Emergencies

In case of road closures or significant traffic, consider:

- Using alternative village roads parallel to the Salem - Kochi Highway.
- Consult local traffic updates for temporary diversions.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Ensure compliance with Tamil Nadu's state regulations, which include:

- Restricted transport times for hazardous materials (typically nighttime).
- Mandatory signage on trucks transporting dangerous goods.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- There have been occasional reports of accidents involving heavy vehicles due to human error or poor road conditions, particularly during the rainy season.
- Few incidents related to hazardous materials noted but primarily caused by improper handling or vehicle faults.



## 8. Environmental Considerations and Sensitive Areas

- The route passes close to agricultural lands which can be sensitive to spills or leakages.
- Villages and small towns along the route require noise and air pollution considerations, especially near schools and hospitals.

## 9. Analysis of Communication Coverage

- Most areas along the route have good mobile network coverage.
- Potential dead zones may exist in more rural stretches, particularly between Padaiveedu and Erumaipatti on local roads.

## 10. Estimated Emergency Response Times

- **Urban Sections:** 20-30 minutes due to better access to hospitals and fire stations.
- **Rural Sections:** Could extend to 45 minutes or more depending on the exact location and time of day.

## 11. An Overall Summary of Risk Assessment

This route entails moderate risk due to varying road conditions, weather impacts, and congestion zones. Drivers should exercise caution during peak hours and in inclement weather. Regular vehicle checks and compliance with local transport laws will enhance safety. It is advisable to ensure effective communication tools and plan for alternative routes when necessary.

The overall assessment highlights the importance of proper planning and awareness to minimize risks, particularly for vehicles transporting hazardous materials.

### Risk Assessment - Turns

|   | Risk Type | Risk Level | Coordinates        | Speed Limit |
|---|-----------|------------|--------------------|-------------|
| 0 | Turn      | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 1 | Turn      | Medium     | 11.43961, 77.87341 | 30 KM/Hr    |
| 2 | Turn      | Medium     | 11.43971, 77.87348 | 30 KM/Hr    |
| 3 | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4 | Turn      | Medium     | 11.44898, 77.87410 | 30 KM/Hr    |
| 5 | Turn      | Medium     | 11.45348, 77.85706 | 30 KM/Hr    |
| 6 | Turn      | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 7 | Turn      | Medium     | 11.45697, 77.82719 | 30 KM/Hr    |
| 8 | Turn      | High       | 11.45542, 77.81725 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 9  | Turn      | High       | 11.45631, 77.81668 | 15 KM/Hr    |
| 10 | Turn      | High       | 11.49409, 77.88004 | 15 KM/Hr    |
| 11 | Turn      | Medium     | 11.49419, 77.88015 | 30 KM/Hr    |
| 12 | Turn      | High       | 11.49383, 77.88442 | 15 KM/Hr    |
| 13 | Turn      | High       | 11.49446, 77.88491 | 15 KM/Hr    |
| 14 | Turn      | Medium     | 11.56083, 77.89921 | 30 KM/Hr    |

## Emergency Locations

|   | type     | name                 | coordinates          | speed_limit | risk_level |
|---|----------|----------------------|----------------------|-------------|------------|
| 1 | hospital | Sri Shanthi Hospital | 11.569272, 77.901937 | 30 km/h     | Medium     |

## Crowded Spots

|   | type   | name                       | coordinates            | speed_limit | risk_level |
|---|--------|----------------------------|------------------------|-------------|------------|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193, 77.8142445 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Turn

**Risk Level:** High



**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**  
**Risk Level: Medium**  
**Speed Limit: 30 KM/Hr**  
**Coordinates: 11.44898, 77.87410**



**Risk Type: Turn**  
**Risk Level: Medium**  
**Speed Limit: 30 KM/Hr**  
**Coordinates: 11.45348, 77.85706**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46318, 77.84913



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45697, 77.82719





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45542, 77.81725



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45631, 77.81668



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49409, 77.88004





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49419, 77.88015



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.49383, 77.88442





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.49446, 77.88491



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56083, 77.89921

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